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Deep Learning based Pulmonary Embolism Detection using Convolutional Feature Maps of CT Pulmonary Angiography Images

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Abstract

Pulmonary Embolism (PE) is a blood clot in the pulmonary arteries of the lungs. Currently, Computerized Tomography Pulmonary Angiography (CTPA) scans are used to diagnose this condition. However, to manually locate the presence of a PE in the scan is laborious. In this paper, the detection of PE and prediction of its features like location, Right Ventricle to Left Ventricle (RV/LV) ratio, and chronicity are automated. First, for the detection of the embolisms, image features from the RSNA Pulmonary Embolism CT (RSPECT) dataset - containing over 12,000 CTPA studies, are extracted by several pre-trained Convolutional Neural Networks (CNN) - AlexNet, Inception V3, ResNet-18, and ResNet-50. Then, various Machine Learning (ML) classifiers are trained on these features and their performances are compared. Second, for the prediction of labels, multi-label classification is performed by training these networks with Binary Cross Entropy Logits (BCE Logits) loss. The results showed that for the detection task, k-NN classifier trained on ResNet-50 features achieved the highest sensitivity of 0.98 with 1.7 false positives per scan and an AUROC score of 97.4%. This system also improved on the state-of-the-art sensitivity score by 0.0975 and AUROC score by 3.4%. For multi-label classification, ResNet-18 performed comparatively better with the best validation loss of 0.07 and weighted macro-average ROC AUC score of 0.61.

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Keywords: Pulmonary embolism; transfer learning; Convolutional Neural Network; Computerized Tomography Pulmonary Angiography images; Deep Learning.

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1. Introduction

Pulmonary Embolism (PE) is a blood clot in the pulmonary arteries of the lungs. It can be life-threatening as it deprives the blocked portions of the lung of blood flow and decreases the oxygen levels. About 115 per 100,000 individuals are diagnosed with PE each year in the US with an overall mortality rate for untreated massive PEs being 18%-65% [Barco, *et al.* (2021)], [Bělohlávek, *et al.* (2013)]. The clot is usually caused by a thrombus travelling up to the lungs from deep veins in the legs (deep vein thrombosis) or occasionally due to the accumulation of fat, or tumours.

As most patients die within the first few hours of getting an embolus, early diagnosis is crucial. Computerized Tomography Pulmonary Angiogram (CTPA) of the thoracic region is the most common front-line diagnostic technique. Each scan consists of up to 128 slices and manually reading each slice and identifying the presence of emboli is laborious and error prone. Further, after diagnosis, the radiologists study several other parameters of the embolism - location, Right Ventricle to Left Ventricle (RV/LV) ratio of the heart and morphological factors, to determine its severity and the prognosis [Ghaye, *et al.* (2006)]. This is a long and tiring task. Thus, Computer-Aided Diagnosis/Detection (CAD) Schemes have been developed to handle this problem more efficiently and accurately. The main motivation/objective for this problem is to develop a deep learning-based system for automatic detection and characterization of pulmonary embolism (PE) from CT pulmonary angiography (CTPA) images.

2. Literature Survey

In recent years, various CAD schemes have been proposed to detect the presence of pulmonary embolisms in Computerized Tomography (CT) images. Most of them follow a three-stage process. Firstly, the possible regions of interest are selected by candidate detection. Secondly, different handcrafted features are extracted from these PE candidates. Finally, simple rule-based methods or machine learning classifiers are used to reduce the false positives.

X. Yang et al. (2019) improved the candidate detection by using a 3D fully convolutional network with skip connections. Vessel aligned 3D cubes are extracted using a 3D location map and fed to a ResNet-18 model for classification. This achieved a detection sensitivity of 84% at 2 FP per study. A U-Net method based on the VGG-16 weights [Wen, *et al.* (2019)] produced an F1 score of 0.8221, recall of 0.7767 and precision of 0.8825.

With the recent advent of artificial intelligence and deep neural networks, systems using these methods have shown a great improvement in the performance metrics. This also owes to the fact that medical imaging data is more readily available. A novel vessel-aligned emboli image representation is proposed by Tajbakhsh, et al. (2015) and a Convolutional Neural Network (CNN) is used to classify the candidates. The initial candidates are detected using a tobogganing algorithm after lung segmentation. The model produced a detection sensitivity of 83% at 2 FP per study.

Finally, the classification of true positive PE candidates is done by using either a simple rule-based classifier [Özkan, et al. (2013a)], [Özkan, et al. (2014)] or by applying various machine learning algorithms like Decision trees, k-NN, support vector machines [Bi and Liang, (2007)], or artificial neural networks [Tang, *et al.* (2010)], [Bouma, et al. (2009)]. The current CAD schemes perform with detection sensitivities ranging from 61% to 90% and False Positives (FP) from 2.6 to 18.4 per examination. The major limitation of these traditional CAD schemes is the high FP rates and inability to capture lower contrast and smaller PE's.

Most methods perform candidate detection by following either a lung segmentation or vessel segmentation scheme. In the former, the lung and mediastinum regions are segmented from the image based on a predefined Hounsfield Unit (HU) intensity range [Bouma, *et al.* (2009)], [Özkan, *et al.* (2013a)]. However, this method does not work well "universally" as the threshold range varies with different CT acquisition techniques. This challenge is overcome by using an adaptive thresholding technique [Park, *et al.* (2011)]. In the vessel segmentation scheme, the pulmonary vessels are segmented by using an adaptive threshold. While the PE candidates are extracted by Bouma et al. (2009) using a novel tracking algorithm to create and apply a binary mask, a tobogganing algorithm is used by others to directly locate the candidate points from emboli properties [Liang and Bi, (2007)], [Park, *et al.* (2011)], [Tajbakhsh, *et al.* (2015)].

From the PE candidates, various intensity, contrast, 3D and 2D shape, size, edge, and boundary-based features are extracted to distinguish between true positive and false-positive candidates. Other features like the shape of vessels and lumen, the location of each candidate and associations between PE using geodesic distance maps are also used to detect True Positive (TP) PE candidates [Liang and Bi, (2007)], [Bi and Liang, (2007)]. An optimal and nonredundant set of features is selected using a genetic algorithm [Park, et al. (2011)] or stepwise regression [Bouma, *et al.* (2009)], [Zhou, *et al.* (2009)] and fed to the classifier.

Transfer learning has reduced the high computational power requirement and the need for a large dataset to handle deep learning problems. Instead of training a CNN afresh, a pre-trained model can be fine-tuned to suit the application

at hand and reduce the training time. In this paper, the performance of different pre-trained models is compared and evaluated for detecting PEs and predicting other labels from CTPA images.

3. Dataset

3.1. RSPECT dataset

The dataset [RSPECT Dataset, (2020)], made available by the Radiologist Society of North America (RSNA) in collaboration with the Society of Thoracic Radiology (STR) consists of over 12,000 CT studies. It is currently the largest annotated pulmonary embolism dataset that is public and is made available as a part of the RSNA-STR PE Detection Challenge 2020. The chest CTPA images were contributed by five international research centers with over 80 expert thoracic radiologists labelling detailed clinical annotations for each. The image data contains Digital Imaging and Communications in Medicine (DICOM) images [Fig 1] that store additional metadata about the study. These images are organized in directories by study and series. There are a total of 17 labels for each image. Fig 2 shows the relationship between various labels. There are three labels that specify the Unique ID (UID) for study, series, and image. The remaining labels are of three categories - ones that are uniquely specified for each image (image-level), ones that characterize a particular image exam or study (exam/study level), and ones that are purely informational and don't require predictions (Table 1). Fig 3 shows how the data fields values are distributed for each label in the dataset. This dataset is readily available in https://www.kaggle.com/c/rsna-str-pulmonary-embolism-detection/data.

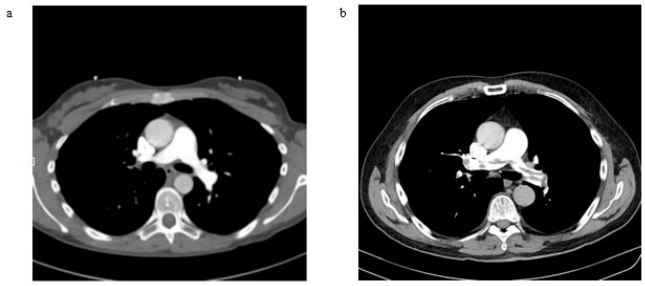


Fig 1. CTPA images containing (a) no pulmonary embolism and (b) multiple pulmonary embolisms.[17]

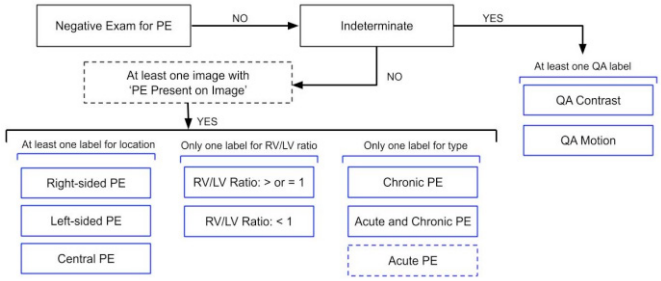


Fig 2. Hierarchy amongst the labels[17]

Table 1. Labels annotated for study.

Image Level	**Exam Level**	**Informational**
Pe_present_on_image	Negative_exam_for_pe Rv_lv_ratio_gte_1 Rv_lv_ratio_lt_1 Leftsided_pe Chronic_pe Rightsided_pe Acute_and_chronic_pe Central_pe Indeterminate	Qa_motion Qa_contrast Flow_artifact True_filling_defect_not_pe

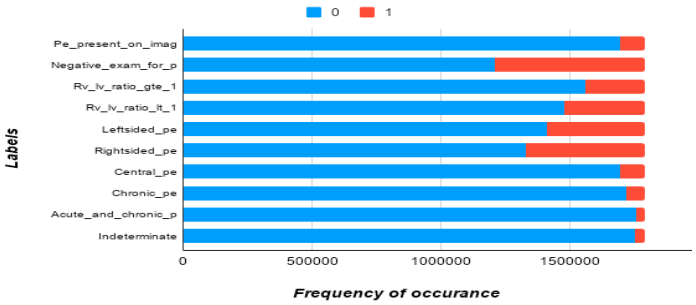


Fig 3. Distribution of data field values

3.2. CTPA images

CT Pulmonary Angiogram is the primary tool for diagnosing a pulmonary embolism. Intravenous contrast dye is administered through veins and the scan is obtained when it is in the pulmonary circuit. The presence of an embolus prevents this contrast dye from filling up the entire blood vessel. This is seen as 'filing defects' that indicate the presence of a PE. The CTPA images are stored in the DICOM format and annotations in a comma-separated values file. Image files are stored in a single folder and named according to the value stored in the SOPInstanceUID DICOM metadata tag, which is a unique identifier for each image. It is an internationally accepted standard to store, exchange and communicate digital medical images. The RSNA Pulmonary Embolism CT (RSPECT) dataset contains 512 x 512 DICOM CTPA images imaged at the arterial phase. The pixel values correlate with the Hounsfield Scale. The Hounsfield unit is used to measure the radiodensity information captured when the tissue is exposed to X-rays. It generally lies in the range of -1000 HU to 1000 HU. Water has 0 HU and different tissue structures have different HU. Hence, windowing is used to selectively display the range of HU where our region of interest is present. In this problem, a lung window (-600 level, 1500 width) is used to view and highlight the lung parenchyma [Wittram, (2004)]. Fig 4 shows the HU distribution of one CT image. We can infer that most of the area is filled with air as air has -1000 HU. The next highest peak is near 0 HU, indicating an abundance of soft tissue, muscle, and some fat.

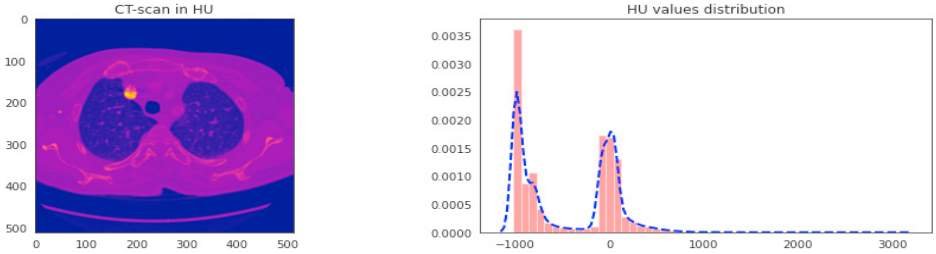


Fig 4. Distribution of HU values in a CTPA image[17]

3.3. Feature labels

3.3.1. Location of the embolus

The location of the embolus is labelled depending on its position in the lungs - 'central', 'right' and 'left'. In radiology, the scans follow an inverted pattern; the right lung is on the left side of the scan and vice versa. A PE is labelled 'right' and 'left' if it occurs in the right and left lungs respectively. A 'central' PE occurs sided, left-sided, and central PEs which are not mutually exclusive, i.e., one image can have location labels enabled.

CT Pulmonary Angiogram is the primary tool for diagnosing a pulmonary embolism. Intravenous contrast dye is administered through veins and the scan is obtained when it is in the pulmonary circuit. The presence of an embolus prevents this contrast dye from filling up the entire blood vessel. This is seen as 'filing defects' that indicate the presence of a PE. The CTPA images are stored in the DICOM format. It is an internationally accepted standard to store, exchange and communicate in the pulmonary trunk region. Fig 5 demonstrates the different location labels. Patients can have multiple PEs across both lungs: hence, we have labels for right-sided, left-sided, and central PEs which are not mutually exclusive, i.e. one image can have location labels enabled.

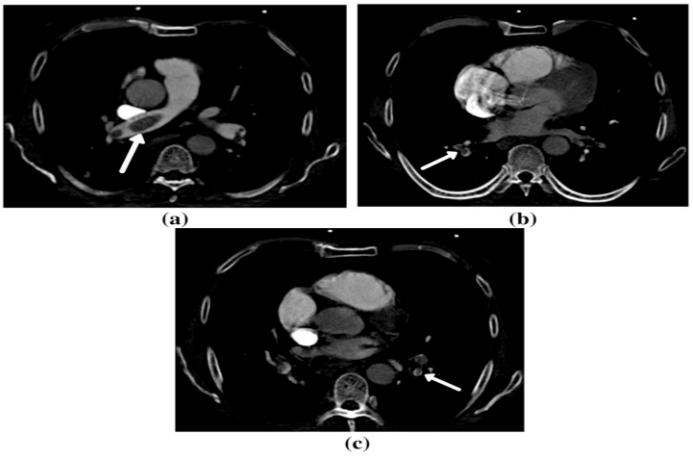


Fig. 5 Different locations of PE (a) Central - PE present in pulmonary trunk, (b) Right -PE present in the right lung, (c) Left - PE present in the left lung viewed using CT Angio window (600/300)[17]

3.3.2. RV dilation

The other important factor to consider while detecting PE is Right Ventricle (RV) dysfunction. It is measured by the ratio of the diameter of the Right Ventricle to that of the Left Ventricle of the heart - RV/LV ratio. Since the left ventricle pumps oxygenated blood to all the organs, under normal conditions, its diameter is more than that of the right ventricle, making the RV/LV ratio < 1. However, when a PE is present, there is more strain on the right ventricle to pump blood to the lungs, and thus, it appears larger than the left ventricle. Hence, an RV/LV ratio >= 1 (Fig 6) indicates the presence of PE (with ratio > 1.1 being extremely severe).

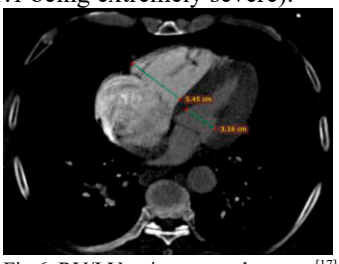


Fig 6. RV/LV ratio greater than one [17]

3.3.3. Severity – acute vs chronic

An acute PE (Fig 7) is characterized by partial filling defects located centrally, forming a polo mint-like sign or by 'railway track' signs on different axes of vessel images. It can also be located eccentrically, and if so, it forms acute angles with the walls of the vessels. The artery with the blockage may also appear enlarged compared to its adjacent patent vessels [Wittram, (2004)].

A chronic PE (Fig 8) leads to a decrease in the vessel diameter due to the complete occlusions. It manifests as irregularities formed as recanalization, bands (ribbon-like structures attached to the walls of vessels at two ends and unattached in the middle), webs (bands forming branches and networks) and flaps. The partial filling defects observed in chronic PE usually form obtuse angles with the vessel walls [Wittram, (2004)].

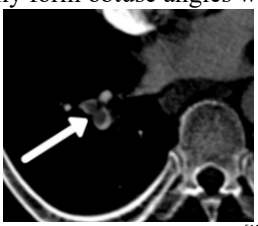


Fig 7. Acute PE (Polo mint sign)[17]

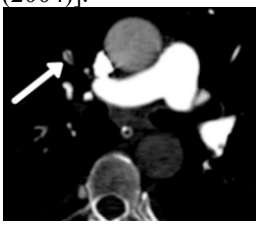


Fig 8. Chronic PE (nonocclusive intraluminal web)[17]

4. Preprocessing

The 512 x 512 DICOM images in the RSPECT dataset converted into 256 x 256 x 3 JPEG images are used [Pan, (2020)]. The 3 channels (RGB) represent the lung, PE, and mediastinal windows. This data is split into a training and validation set with 6500 and 779 studies respectively. These training images are selected at random and are transformed by applying brightness contrasts, horizontal and vertical flips, and downscaling. This is done to incorporate randomness in the data and to avoid overfitting. Further, these transformed images are normalized, centred, and standardized with mean as [0.485, 0.456, 0.406] and the standard deviation as [0.229, 0.224, 0.225]. Before feeding these images into the models, both the training and validation images are resized according to the input requirements of the pre-trained models. Pre-processing can be achieved by using of the software provided by https://www.rsna.org/-/media/Files/RSNA/Covid-19/RICORD/RSNA-Anonymizer-Program-Instructions.pdf

5. Proposed Model

In this paper, we propose a two-stage classifier for the detection of PE and their corresponding feature labels. As the model uses Deep Learning techniques, the model automatically takes care of the feature extraction. The first stage consists of detecting if the image and the associated study contains an embolism while the second stage is responsible for predicting the feature labels discussed in the previous sections. The workflow is demonstrated in Fig 9.

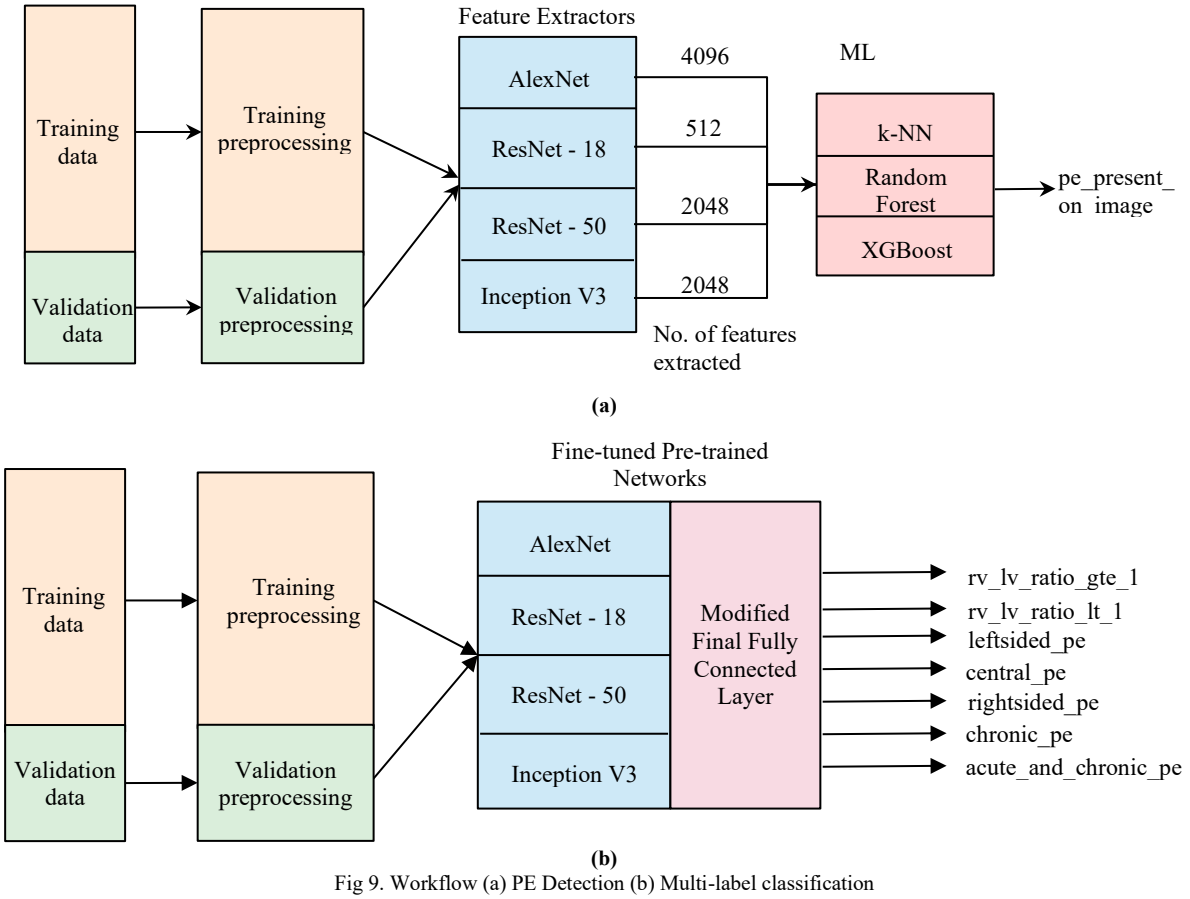


Fig 9. Workflow (a) PE Detection (b) Multi-label classification

AlexNet employs convolutional operations with learnable filters, employing operations like convolution, pooling, and fully connected layers. The model utilizes softmax regression for classification, where it computes the probability distribution over classes. ResNets is the residual learning framework, where instead of learning the actual mapping, the network learns the residual (difference) between the input and the output. The Inception module uses parallel convolutions of different kernel sizes, allowing the network to capture information at various scales. It employs 1x1, 3x3, and 5x5 convolutions along with pooling operations. The output from these parallel operations is concatenated and fed into subsequent layers

5.1. PE detection

To detect if an image has an embolus present, four CNN models - AlexNet [Krizhevsky, et al. (2012)], ResNet-18], ResNet-50 [Szegedy, et al. (2016)], and Inception V3 [He, et al. (2016)] are used as feature extractors. These models are pre-trained on the ImageNet dataset and have assigned weights. For each network, the classifier, i.e, the final fully connected layer is removed. Instead, the input to this layer is forwarded to the output and stored. These are the feature values extracted by the model for each image. Various classifiers like k-Nearest Neighbour (k-NN), Random Forest and XGBoost are trained on these features to predict the presence or absence of PE in the images. k-NN performed best with the number of nearest neighbours set to 5, Random Forest with 100 estimators, and XGBoost fast histogram algorithm with 0.1 learning rate and subsample set to 0.69. The workflow is demonstrated in Fig 9 (a).

5.2. Multi-label classification

Table 2 shows the various pre-trained networks, the number of convolutional layers present, the number of learnable parameters, and their implementation details. The final fully connected layer is modified into a new linear layer to output probabilities for seven class labels. Based on the memory and time constraint, the best batch size is determined, and the models are trained for ten epochs. The Adam optimizer is used with a weight decay of 0.00001. The learning rate is initially set to 5e-4 and is dynamically adjusted using the cosine annealing schedule. The loss function used here is Binary Cross

Entropy with Logits Loss (BCELogits) as all the labels have binary values. The BCELogits loss has better numerical stability than the binary cross entropy loss function. The BCELogits Loss function is defined as

$$BCELogits(p(y), y) = -w_n[\, y_n \cdot \log t + (1 - y_n) \cdot (1 - \log t)] \tag{1}$$

where $t = \sigma(p(y))$, y is the actual value, p(y) is the probability predicted by the network, and w_n is the weight vector for the nth sample.

Table 2. Description of the pre-trained networks and their implementation details.

Pre-trained model	# of Convolutional layers	# of learnable parameters	Training batch size	Input image size
AlexNet	5	57M	128	224 x 224x 3
ResNet-18	17	11M	128	224 x 224 x 3
ResNet-50	48	23M	128	224 x 224 x 3
Inception V3	48	24M	32	299 x 299 x 3

Table 3. Various performance metrics of different classifiers

Feature Extractor	Classifier	Accuracy	Precision	Recall	F1 Score	ROC AUC Score
AlexNet	XGBoost	0.834	0.84	0.83	0.81	0.704
	k-NN	0.78	0.77	0.78	0.78	0.704
	Random Forest	0.83	0.84	0.83	0.81	0.704
ResNet-50	XGBoost	0.883	0.88	0.88	0.88	0.822
	k-NN	0.981	0.98	0.98	0.98	0.974
	Random Forest	0.864	0.87	0.86	0.85	0.77
ResNet-18	XGBoost	0.865	0.87	0.87	0.86	0.777
	k-NN	0.937	0.94	0.94	0.94	0.919
	Random Forest	0.841	0.86	0.84	0.82	0.709
Inception v3	XGBoost	0.835	0.86	0.83	0.81	0.686
	k-NN	0.676	0.61	0.68	0.61	0.514
	Random Forest	0.812	0.85	0.81	0.78	0.686

6. Result

The optimal hyperparameters are obtained from exhaustive grid search algorithm.

Table 4. Hyperparameter of different models

Hyperparameter	AlexNet	ResNet-18	ResNet-50	Inception V3
Optimizer	SGD	SGD	SGD	SGD
Momentum	0.9	0.9	0.9	0.9
Weight decay	0.005	0.0001	0.0001	0.0001
Batch size	256	256	256	256
Epochs	90	100	100	300
Image size	224x224	224x224	224x224	299x299

6.1. Detection of PE

Table 3 shows the performance of the various classifiers for the 4 feature extractors. The metrics that are used to compare the results include accuracy, precision, recall, F1 score and AUC ROC score. As it can be observed, the k-NN classifier trained on the feature values extracted by ResNet-50 outperformed the other classifiers in all metrics. This system achieved the highest sensitivity of 98% with 1.7 false positives per scan. It also has the maximum ROC AUC Score indicating a 97.4% chance that the model will be able to differentiate the presence or absence of a PE.

6.2. Feature label prediction

The training and validation losses of the models through the epochs trained are shown in Fig 10. ResNet-18 was able to achieve the best validation loss of 0.07. The higher training loss of inception can be attributed to the inclusion of the dropout layer while training and exclusion during validation. Fig 11. shows the ROC curve of the models for each label. Since the dataset is normalized to include an equal ratio of positive and negative classes, the ROC AUC score is used to compare the model performances. For the prediction of RV/LV ratio and location labels, the ResNet-18 model, and for chronicity labels, ResNet-18 and ResNet-50 models revealed better scores. To compare the models on multi-label classification, the weighted macro-average ROC AUC score is used, and ResNet-18 has the highest AUC of 0.61.

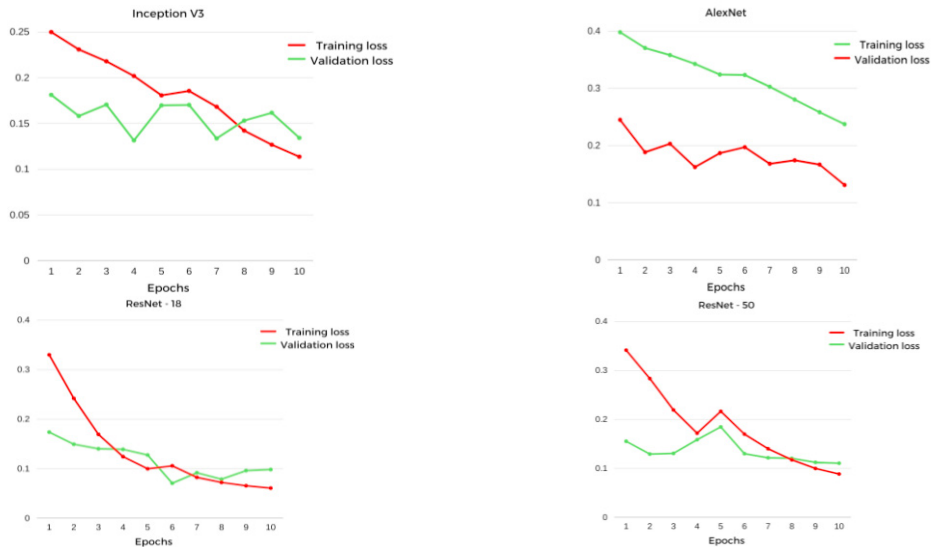


Fig 10. Training and validation loss through epochs of the models

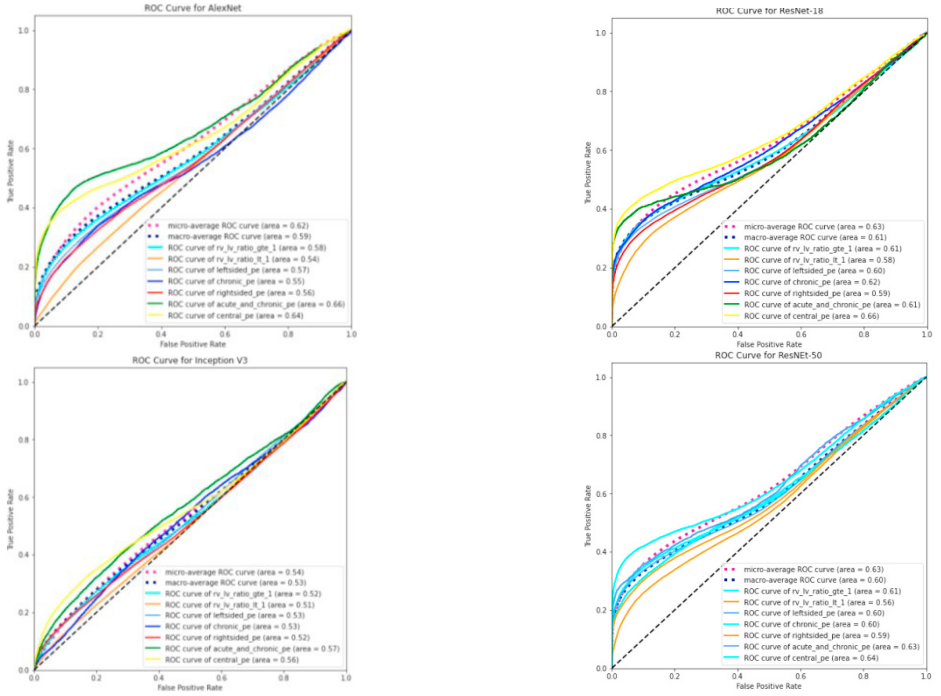


Fig 11. ROC Curve of the four models.

7. Discussion and Conclusion

In this paper, the performance of various pre-trained models on the detection of PE and multi-label classification has been compared and evaluated. The results show that the k-NN classifier on ResNet-50 features outperforms the state-of-the-art AUROC score by 0.034 [Rajan, et al. (2020)], sensitivity score by 0.0975 with only 1.7 FP per study [X. Yang, et al. (2019)]. While the pre-trained models worked well with the detection task, the results aren't consequential with the multi-label classification. This can be attributed to the class imbalance problem. While this issue was resolved by random sampling in PE detection, it wasn't practical to do so for all other labels individually. The advantage of using the feature extraction was the reduced training time and better classification. Although k-NN took longer to classify than the other ML classifiers, it was well within the available GPU and memory resources, which can be traded off for performance. The limitation of this study is the minimal number of epochs for training owing to the limited resources. Further, the dataset consists of slices with 2.5 or 3mm thickness, which can oversee smaller and insubstantial PEs. In the future, the models can be compared with better resources and the multi-label classification can be improved. The performance on the detection of PE can be further improved through the utilization of vision transformer(VIT) model. In conclusion, this work aims to solve two tasks. First, to strategically use various pre-trained models as feature extractors and classifiers to detect PE. Second, to perform multi-label classification using these models and compare the results obtained. The method for PE detection provides a basis for reduced training times and accurate detection aiding any future study in this direction.

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